Random Forests

Random forests are an ensemble learning technique that combines multiple decision trees to improve accuracy and robustness. By aggregating the predictions of several trees, random forests mitigate overfitting and enhance generalization.

How Random Forests Work:

1. **Bootstrapping:** Random subsets of the data are sampled with replacement to train each tree.
2. **Feature Randomness:** A random subset of features is used for splitting at each node, ensuring diverse trees.
3. **Aggregation:** Predictions are averaged (regression) or determined by majority vote (classification).

Advantages of Random Forests:

- **Reduced Overfitting:** By combining multiple trees, random forests are less prone to overfitting compared to individual decision trees.
- **High Dimensionality:** They handle datasets with a large number of features effectively.
- **Feature Importance:** Random forests provide insights into the relative importance of features.

Random Forest Implementation in Python

Below is an example of implementing a random forest classifier: from sklearn.ensemble import RandomForestClassifier # Train the model

```
random_forest = RandomForestClassifier(n_estimators=100, random_state=42)
random_forest.fit(X_train, y_train)

# Predictions
predictions = random_forest.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, predictions)
print("Accuracy:", accuracy)
```

Insights